

Asim Waheed

Waterloo, ON, Canada | asim.waheed29@gmail.com
asimwaheed.com | [GitHub](#) | [LinkedIn](#) | [OCRID:0009-0001-9618-1082](#)

Education

PhD in Computer Science

Sep 2025 – Present

University of Waterloo, Canada

Supervisors: Florian Kerschbaum, N. Asokan

M.Math in Computer Science

Sep 2021 – Aug 2023

University of Waterloo, Canada

Supervisor: N. Asokan

Thesis: On Using Embeddings for Ownership Verification of Graph Neural Networks

BS in Computer Science

Aug 2016 – May 2020

Lahore University of Management Sciences, Pakistan

Experience

Doctoral Researcher

Sep 2025 – Present

University of Waterloo, Canada

Research Engineer

Oct 2023 – Aug 2025

University of Waterloo, Canada

- Developed **Amulet**, a framework for trustworthy ML (technical report).
- Designed an Intel SGX-based framework for secure attestation of ML model properties.

Graduate Student Researcher

Sep 2021 – Aug 2023

University of Waterloo, Canada

- Developed a defense against model-stealing attacks on graph neural networks (IEEE S&P 2024).
- Contributed to an NLP-based auditing framework for Android APIs (USENIX Security 2023).

Data Scientist

Sep 2020 – Aug 2021

Data Science Dojo, USA (Remote)

- Led enterprise AI consulting projects using Azure Cognitive Services.
- Built production analytics and NLP systems.

Research Assistant

Sep 2019 – Aug 2020

Lahore University of Management Sciences, Pakistan

- Conducted research on backdoor defenses in NLP models (USENIX Security 2021).
- Supported faculty research through infrastructure management and data pipeline development.

Machine Learning Engineer Intern

Jun 2018 – Aug 2018

Paitoo (Food Technology Start-up), Pakistan

- Built text classification and recommendation systems.
- Deployed backend services using Docker and Kubernetes.

Publications

1. Vasisht Duddu, Lipeng He, **Asim Waheed**, N. Asokan. *SoK: Colluding Adversaries in Machine Learning Pipelines*. USENIX Security 2026. **Proceedings**.
2. **Asim Waheed**, Vasisht Duddu, Rui Zhang, Sebastian Szyller. *Amulet: A Python Library for Assessing Interactions Among ML Defenses and Risks*. arXiv preprint, 2025. [arXiv:2509.12386](#).
3. **Asim Waheed**, Vasisht Duddu, N. Asokan. *GrOVe: Ownership Verification of Graph Neural Networks using Embeddings*. IEEE S&P 2024. **Proceedings** [arXiv:2304.08566](#).
4. Parjanya Vyas, **Asim Waheed**, Yousra Aafer, N. Asokan. *Auditing Framework APIs via Inferred App-side Security Specifications*. USENIX Security 2023. **Proceedings**.
5. **Asim Waheed**. *On Using Embeddings for Ownership Verification of Graph Neural Networks*. Master's Thesis, University of Waterloo, 2023. [UWSpace](#).
6. **Asim Waheed**, Sara Qunaibi, Diogo Barradas, Zachary Weinberg. *Darwin's Theory of Censorship: Analysing the Evolution of Censored Topics with Dynamic Topic Models*. WPES 2022. **Proceedings**.
7. Ahmadreza Azizi, Ibrahim Asadullah Tahmid, **Asim Waheed**, Neal Mangaokar, Jiameng Pu, Mobin Javed, Chandan K. Reddy, Bimal Viswanath. *T-Miner: A Generative Approach to Defend Against Trojan Attacks on DNN-based Text Classification*. USENIX Security 2021. **Proceedings** [arXiv:2103.04264](#).

Teaching & Mentoring**Teaching Assistant**

2021 – 2023

University of Waterloo

- CS458: Computer Security and Privacy
- CS480: Introduction to Machine Learning
- Multiple intro courses (CS105, CS135, CS246)

Teaching Assistant

2018 – 2020

Lahore University of Management Sciences

- CS437: Deep Learning
- CS334: Principles and Techniques of Data Science

Mentor

Oct 2021 – Dec 2021

GradApp Lab (voluntary)

- Advised undergraduate students on graduate school applications and career planning; mentee admitted to the University of Toronto and currently employed as a Data Scientist with the Ontario Government.

Academic Service

- **Artifact Evaluation Committee**, Privacy Enhancing Technologies Symposium (PoPETS) 2026

Scholarships & Awards

- **David R. Cheriton Graduate Scholarship**, University of Waterloo (PhD) 2025–2027
- **David R. Cheriton Graduate Scholarship**, University of Waterloo (M.Math) 2021–2023
- **International Master’s Award of Excellence**, University of Waterloo (MMath) 2021–2023
- **Graduated with High Distinction**, Lahore University of Management Sciences 2020
- **Dean’s Honor List (all four years)**, Lahore University of Management Sciences 2016–2020